

Comparing Global Thresholding and Random Forest Classification for Improved Flood Mapping Using Sentinel-1 Data: A Case Study of the 2023 Kakhovka Dam Breach

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ABSTRACT

On 6 June 2023, the destruction of the Kakhovka Dam in southern Ukraine caused extensive flooding across the Kherson region. During such events, rapid and reliable flood mapping is essential for emergency response and impact assessment. This study compares a traditional Global Threshold (GT) approach based on Otsu's method with a supervised Random Forest (RF) classifier for mapping flood extent using Sentinel-1 data. Pre- and during-flood Sentinel-1 VV and VH images acquired on 2 and 9 June 2023 were processed and analysed. A flood map developed using high-resolution (3m) Planet Labs satellite imagery was used as reference data. Flooded and non-flooded areas were quantified for each method and compared to the reference map using area-based metrics. The GT method significantly overestimated the flooded area, with a high commission error of 1,143 km². This was due its misclassification of wet sandy surfaces and some agricultural land as flooded. Conversely, the RF classifier produced estimates closer to the reference data, with a lower commission error of 67 km². However, the RF model showed an omission error of 251 km², primarily in vegetated areas where C-band waves could not fully penetrate canopies, and in built-up areas where double-bounce effects maintained high VV backscatter despite flooding. While the GT method is faster to implement and can be useful for rapid assessment, the results show that the RF approach provides a more reliable representation of flood extent. The findings highlight the potential of supervised machine-learning for improved flood mapping using multi-temporal Sentinel-1 data.

KEYWORDS: Flood Mapping; Global Thresholding; Kakhovka Dam; Random Forest; Sentinel-1 SAR;

1 INTRODUCTION

On 6 June 2023, the Kakhovka Dam in Kherson, Ukraine, was destroyed during the armed conflict between Ukraine and Russia (Vyshnevskiy, et al., 2023). This caused a sudden release of water from one of the largest reservoirs in Europe (Yailymov, et al., 2025). According to Yailymov et al. (2025), more than 600 km² were flooded within three days, affecting over 700,000 people and causing extensive damage and at least 58 fatalities. The event led to an estimated \$148 billion in overall damages and losses (Yailymov, et al., 2025). During flood events with such widespread impacts, accurate and timely mapping of flood extent is essential for emergency response and damage assessment.

Optical satellite imagery is often limited during floods due to cloud cover and time restrictions. Synthetic Aperture Radar (SAR) overcomes these limitations, as it can operate in all weather conditions and at night (Liang & Liu, 2020). However, producing precise flood maps from SAR data, specifically from Sentinel-1, remains difficult. Traditional mapping often relies on global thresholding (GT), such as that proposed by Otsu (1979). This method identifies a single optimal threshold by maximising the between-class variance to separate water from land across an entire scene (Liang & Liu, 2020). However, the problem with this "one-size-fits-all" approach is that it does not account for the heterogeneous land surfaces, such as urban areas, forests, and agricultural fields (Liang & Liu, 2020). As a result, smooth land features like roads, dry soil or sand can be easily misclassified as water.

Machine learning methods offer an alternative approach. Random Forest (RF) is a supervised classification algorithm that uses multiple decision trees trained on subsets of the data. It then assigns classes based on majority voting (Billah, et al., 2023). RF classifiers can use multiple input features and capture non-linear relationships, making them suitable for complex classification tasks. When applied to multi-temporal Sentinel-1 data, RF models can learn changes in backscatter

associated with flooding rather than relying on a single threshold value.

This study aims to determine if a supervised RF classifier can improve flood mapping accuracy compared to the Otsu (1979) global threshold method. By comparing the results of these two methods against a reference flood map generated using high-resolution (3m) Planet Labs satellite imagery, this paper evaluates which technique is more reliable for flood extent mapping in heterogeneous landscapes, such as the Kherson area.

2 METHODS

2.1 Data Acquisition and Pre-processing

The study area covers approximately 6,041 km² within and around the Kherson region of Southern Ukraine. Two Sentinel-1 Ground Range Detected (GRD) images were accessed via Google Earth Engine (GEE). These involve a pre-flood and during-flood images acquired on 2 June 2023 and on 9 June 2023 respectively. These images were captured in Interferometric Wide (IW) mode with dual-polarisation (VV and VH). Pre-processing of the images included radiometric calibration, to convert DN to backscatter intensity (σ^0), and correction of topographical distortions using the SRTM 30m DEM. Also, to reduce the characteristic radar speckle, a Refined Lee filter was applied to both images to ensure a smoother signal for classification. The final images obtained after pre-processing are shown in Figure 1.

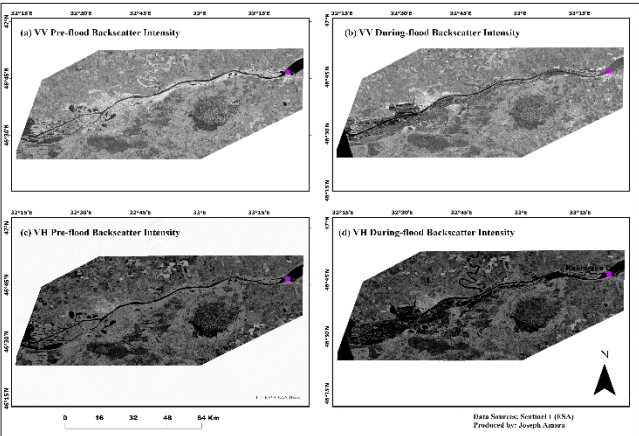


Figure 1: Processed Sentinel-1 backscatter intensity images, showing VV backscatter before (a) and during (b)- flood and VH backscatter before (c) and during (d)-flood. With bright and dark areas respectively indicating high and low backscatter intensities. The images show considerable decrease in backscatter intensity from pre-flood to during-flood (especially evident for VV). The purple dot in all the images is the location of the Dam.

2.2 Reference Flood Data and Permanent Water Bodies

A flood map created by Reiche, et al., (2024) using Planet Labs’ 3m resolution satellite imagery was used as reference for accuracy assessment. This reference map was resampled to a 10m spatial resolution to allow for a direct, pixel-to-pixel comparison with the processed Sentinel-1 imagery. Permanent water bodies were excluded from the analysis using the reference flood map to ensure that the results specifically represented the flood extent caused by the dam breach.

2.3 Global Thresholding (Otsu’s Method)

A global thresholding baseline was established using Otsu (1979) method. This unsupervised technique identifies an

optimal threshold k^* by maximising the between-class variance (σ^2_B) of the backscatter histogram. The optimal threshold k^* is given by:

$$\sigma^2_B(k^*) = \max_{(1 \leq k < L)} \left[\frac{(\mu_T \omega(k) - \mu(k))^2}{\omega(k)(1 - \omega(k))} \right] \dots\dots\dots(1)$$

Where $\omega(k)$ represents the proportional area of the pixels below the threshold, $\mu(k)$ is the mean backscatter intensity of those pixels up to the threshold k , μ_T is the total mean backscatter intensity of the whole image and L represents the total number of discrete grey levels (or histogram bins) used to represent the backscatter distribution. Separate thresholds were calculated for the VV and VH bands on the during-flood image. Areas with backscatter intensities less than both thresholds were classified as flooded and those above either threshold as non-flooded.

2.4 Random Forest Classification

The RF model was configured with 100 decision trees using a multi-temporal stack of VV and VH bands from both the pre-flood and during-flood images in GEE. This resulted in a total of four input bands. This allowed the classifier to detect change over time, which is often a more reliable indicator of flooding than absolute backscatter intensity. The model was trained using a total of 532 samples derived from the reference map. An equal number of samples were collected from flooded and non-flooded areas.

2.5 Map Comparison and Area-Based Evaluation

The performance of each method was evaluated through a direct map comparison against the reference map in ArcGIS Pro. The total flooded and non-flooded area in km² was calculated for each map. The commission and omission errors for both flooded and non-flooded areas were obtained for the GT and RF model. This approach allowed for a clear quantification of the differences in the mapped areas by each method.

3 RESULTS

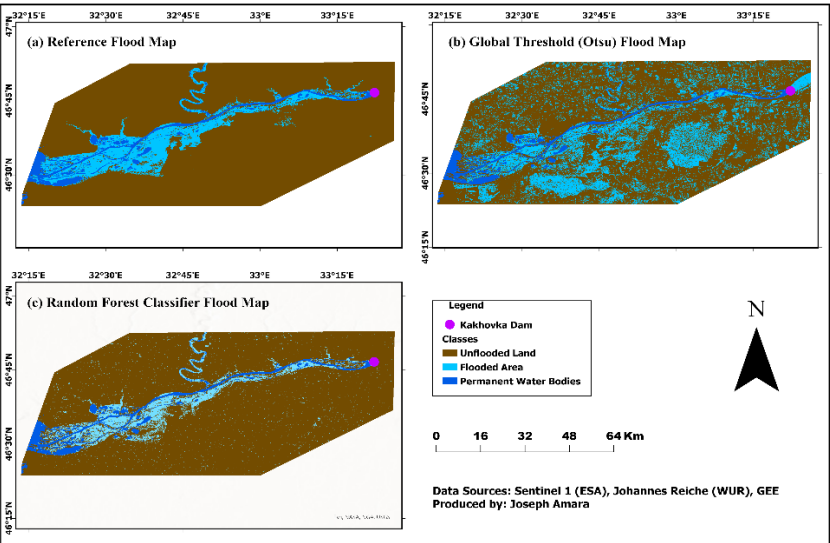


Figure 2: Comparison of flood extent maps for the Kherson region on 9 June 2023: (a) Planet Lab-based reference flood map (3m resolution resampled to 10m); (b) Global (Otsu) Threshold flood map, using VV and VH thresholds of -12.92 dB and -18.96 dB respectively, showing significant overestimation of flooded areas; (c) Random Forest Classifier flood map showing slight underestimation flooded areas, with some speckled flooded areas in the mostly unflooded areas. The location of the Kakhovka Dam is highlighted in all three maps with purple dots.

Table 1: Comparative area-based evaluation of flooded and non-flooded areas derived from the reference, GT, and RF maps; shows the Commission and Omission errors in km² calculated relative to the reference data for each method for flooded and non-flooded. Permanent water bodies were masked using the reference map and therefore remained unchanged.

Class	Reference Area (km ²)	GT Area (km ²)	GT Commission Error (km ²)	GT Omission Error (km ²)	RF Area (km ²)	RF Commission Error (km ²)	RF Omission Error (km ²)
Flooded	675	1,553	1,143	265	491	67	251
Non-flooded	5,075	4,197	265	1,143	5,259	251	67
Permanent Water	291	291	-	-	291	-	-
Total Study Area	6,041	6,041	-	-	6,041	-	-

Figure 2 presents the flood extent maps for the Kherson region on 9 June 2023 derived from the reference map, the GT (Otsu) method, and the RF classifier. Clear differences in the spatial extent of flooded areas are observed between the three maps. The quantitative comparison of flooded and non-flooded areas for each method is summarised in Table 1. The reference flood map indicates a total flooded area of 675 km². In comparison, the GT method, producing VV and VH thresholds of -12.92 dB and -18.96 dB respectively, classified 1,553 km² as flooded, with 1,143 km² of those being unflooded areas wrongly classified as flooded. The method also classified 265 km² of flooded areas as unflooded. The RF classifier identified a flooded area of 491 km², of which only 67 km² unflooded land was wrongly classified as flooded. However, this method also wrongly classified 251 km² of flooded areas as non-flooded.

The area of permanent water bodies remained constant across all maps at 291 km², as these areas were excluded from the flood extent analysis. The total mapped area was consistent across all datasets and matched the defined study area of approximately 6,041 km².

4 DISCUSSION

The comparison between the GT and RF methods reveals significant differences in their ability to map floods accurately. While the GT approach is much faster because it simply calculates a mathematical threshold to separate water from land, its speed comes at the cost of accuracy. In contrast, the RF method had a longer processing time, and required the collection of representative training samples, which increased the overall time. However, it delivered a more reliable result. The GT method's main issue was a massive overestimation of the flood extent, with a commission error of flooded area of 1,143 km². This error was most visible in the southern and south-eastern parts of the study area. These regions contain many sandy surfaces which produce lower radar backscatter due to increased specular reflection (Liang & Liu, 2020). Because the GT method relies on a single "one-size-fits-all" threshold for each polarisation, it wrongly classified these sandy patches as water. Some unflooded agricultural fields were also classified as flooded.

The RF model performed better because it used four bands (VV and VH from both the pre- and during-flood images) to detect changes over time rather than just looking at a single point in time. This allowed it to ignore the sandy areas that

had lower but minor backscatter changes. This caused it to focus on areas with major reduction in backscatter values, indicating actual inundation. However, the RF model did suffer from an omission error of flooded areas of 251 km². Most of these missed flooded areas showed very little or increased backscatter change between the pre- and during-flood images. These areas included highly vegetated areas and built-up areas. For the highly vegetated that were flooded but not completely covered by water, the radar recorded similar VH and VV backscatter values both before and during the flood. This is due to the lack of penetration of the Sentinel-1 C-band (~ 5.6 cm) through the thick vegetation canopies. Some of the flooded built-up areas were misclassified mainly due to increase VV backscatter values resulting from increased double-bounce effect during the flood. Since the RF model relied mostly on identifying decrease in backscatter values, it could not correctly identified flooded areas with minimal changes in pre- and during-flood backscatter values.

5 CONCLUSION

This study compared a GT (Otsu) approach and a RF classifier for mapping flood extent following the June 2023 Kakhovka Dam breach using Sentinel-1 SAR data. The results show clear differences between the two methods. The GT approach substantially overestimated the flooded area, mainly due to its reliance on a single backscatter threshold that could not account for variations in surface characteristics. In contrast, the RF classifier produced flood extent estimates closer to the reference data, with significantly lower commission errors.

The improved performance of the RF method is mainly due to its use of multi-temporal VV and VH data, which allowed it to detect changes in backscatter associated with flooding rather than relying on absolute values. However, the RF approach still underestimated flooding in some vegetated and built-up areas, highlighting its limitations when there is limited backscatter value change.

Overall, the findings indicate that while threshold-based techniques are simple and less computationally intensive, their reliability is limited in heterogeneous environments, such as the Kherson region. This is similar to the findings by Liang & Liu (2020). The study also shows that with representative training data, machine-learning-based approaches, such as RF, can be more suitable for flood mapping in heterogeneous landscapes, than global thresholding methods, such as that proposed by Otsu (1979).

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